

5-Theory - HW

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Date

① Encrypt Rail Fence

PT = Enjoy your Journey" d=2,3

d=2
 E J Y O R O R E
 N O Y U J U N Y

CT = EJYORDENOYUJUNY

d=3
 E N O Y U R J U R N Y^x
 J O O O E

CT = EYR RXNOYUJUNYJOOE

② message : Print Screen

d=2

P I T C E N
 R N S R E

CT = PITCENRNSRE

d=3

P T E
 R N S R E
 I C N

CT = PTE RNSRE I C N

③ Decrypt : E0SSNYUWLEJTYREASON'S

d=4

L=19

19/4=4

E						O						C				S
	N					Y		U				W		L		E
		J		Y				R		E				A		S
			D						N						S	

PT = ENJOY YOUR NEW CLASSES

④ Decrypt BOOIQTURBSNUUTYPRAS $d=3$

B	O	O	I	Q	T	U	R	B	S	N	U	U	T	Y	P	R	A	S
M	Y	U	R	B	O	S	N	U	U	T	Y	P	R	A	S			

PT = BUY YOUR BOOKS IN AUGUST

⑤ SEMSER TISINCMOO $d=2$

S	E	M	S	E	R	T	I
S	I	N	C	M	O	O	

PT = SS EIMNS EEMROTOI

Permutation:

① message = we are discovered key = 53124

5	3	1	2	4
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

CT = ASR RCE EIE EODWDM

DES

① IP in hexadecimal: 0x0002 0000 0000 0001
Initial permutation is given (table)

0000	0000	0000	0010	bit	position
0010	0000	0000	0000	15	→ 63
0000	0000	0000	0000	64	→ 75
0000	0000	0000	0001		

Apply initial permutation

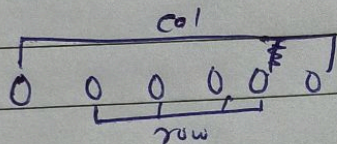
0000 0000 0000 0000 0000 0000 1000 0000
 0000 0000 0000 0000 0000 0000 0000 0000

Hexa : 0x0000 0000 0000 0000

② 4 bit PT : $m = 1101$ $K = 1011$ $\rightarrow XOR$

$$\begin{array}{r} 1101 \\ \oplus 1011 \\ \hline 0110 \end{array} \quad CT = \underline{\underline{0110}}$$

③ $1/p \rightarrow Sbox1 = 000000$



$c=0$ $r=0$ in $Sbox1 = 14$

o/p = 01110